

A Study of the Effects of Network Latency on Visual Task Performance in Video Conferencing

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ABSTRACT

We study the effects of network latency and jitter on visual task performance between participants in video conferencing. Users of video conferencing tools commonly report a phenomenon known as Zoom fatigue. We postulate that this is a consequence of the effort required to maintain visual communication in the face of network latency and jitter. Understanding the relationship between network latency, jitter, and Zoom fatigue can inform the design of visual computing methods to mitigate the effects. To study the causes of Zoom fatigue, we analyzed a visual communication task via video conferencing through a user study employing a game of Charades since it relies on highly visual communication. Our findings demonstrate a direct correlation between user performance while playing the game and network latency and that the level of user engagement in the game plays a vital role in indicating adaptation to the latency when user engagement levels are high.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in collaborative and social computing.**

KEYWORDS

Network Latency, Visual Task Performance, Visual Communication, Zoom Fatigue

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1 INTRODUCTION

Video communication software has seen an exponential rise in use as a result of the COVID-19 pandemic. People are now heavily reliant on video conferencing applications including Zoom, Google Meet and Microsoft Teams. Video conferencing is now being used for everything ranging from online education, health and fitness, doctor's appointments to remote business work. Zoom has emerged as one of the most widely used video conferencing applications and continues to see an ongoing surge in the number of active users [21]. The existence of this and similar applications, have indeed facilitated remote work on a scale not seen before the pandemic. Video-conferencing has also played an important role in enabling people to connect socially, even while they have had to remain physically distant, and helped people to maintain their physical and mental well-being [24]. These technologies have also emerged as a tool for environmental benefit, reducing travel related emissions, saving time and money [5], and discussed as a tool for improving access to conferences and events for people with disabilities [20].

However, video conferencing tools also bring several disadvantages, some of which include prolonged screen-time resulting in negative physiological and psychological effects including: tiredness, anxiety, misunderstandings, stress and/or worry. Researchers now refer to these effects of prolonged use of video conferencing as Zoom Fatigue [30]. Indeed, there is now a recent and growing body of research on this topic [24, 28]. However, despite recognizing the phenomenon, there is still relatively little understanding of the reasons why people find video communication more tiring than face-to-face communication. Understanding the relationship between network conditions and Zoom fatigue can provide insights to design visual computing solutions to mitigate the effects of prolonged Zoom use.

We study the effects of network latency on a highly visual task performed via video conferencing. We analyze visual communication over video conferencing through an experiment employing a game of Charades conducted via the Zoom videoconferencing tool. Charades relies on highly visual communication to convey information via actions to other players. During the game play, we inject increasing levels of latency and jitter to study the impact of this on participants' ability to guess what the other player is communicating. This is in contrast to other studies that focus on audio only and audio-video communication. Our inductive analysis of user

perceptions around the task is used to study the additional cognitive load on participants attributed to network delay and instability. Through this study, we establish that in a video call:

- Participants are able to compensate for latency and jitter in tasks that they find enjoyable, and
- Greater levels of latency and jitter increases irritability, tiredness, reduces the perceived fun of the task and reduces tolerance to latency.

2 RELATED WORK

This research spans work in network communication, collaborative working, visual computing, and zoom fatigue. Here we discuss the relevant prior works in these related fields to explain the context of our research.

2.1 Network Latency and Jitter

Network latency is defined as the time taken to send information in packets from a source on the internet, to a destination. Jitter is the variation in this latency. These characteristics of networks, exist to varying levels in all network communications. In video calls, we experience the effects through; dropped calls and pauses, poor audio or video quality, inability to understand visual cues, distracting visual artifacts or delayed/disjoint video and audio [4]. High-speed networking will improve video conferencing, but fundamental physical limits on network data transmission mean it is impossible to entirely eliminate latency and jitter [2, 22].

Compared to face-to-face interaction, video communication results in longer conversational turns; fewer interruptions, overlaps, greater use of back-channels such as textual chat, and increased formality when switching speakers. Communication on platforms with broadcast quality audio and video can be effective, but they fail to adequately replicate non-verbal communication inherent in face-to-face interaction. [8, 25, 26]. These limitations are an important barrier to collaboration.

2.2 Visual Collaborative Tasks

Non-verbal cues establish a shared visual language and mutual understanding between people considered important for collaboration. These cues help people to clearly express their attitudes [13]. Some non-verbal cues are communicated through video and prior research has shown their importance in collaborative tasks, by comparing these across video and audio-only platforms [18, 25, 26]. Video mediated communication has also been shown to better facilitate conflict resolution and interactive activities [6]. In contrast, other studies demonstrate that video communication can be worse than audio-only conversations [7, 19]. This is attributed to ambiguities in managing naturalistic turn-taking. Latency and jitter cause non-verbal feedback to lose causal association with natural variations in speech. Early studies have also predicted that prolonged exposure to this medium will improve its usability through adaptation [9].

2.3 Zoom Fatigue

Recent research now suggests that online conferences, meetings and events are a poor substitute for "face-to-face". During real-time communication, the human sensory system takes input from

the entire body of others in a shared dialogue to understand what they are communicating. In contrast, video conferencing displays a single frame predominantly covered with the other person's face. This can be perceived at a low (evolutionary) level as threatening. The brain registers the other person's enlarged face, flooding the human body with stress hormones [30]. Several video conferencing platforms reflect only the speaker's frame, eliminating all other participants who are on the call. This makes it difficult to tell if others are still active and whether or not they are able to understand the active speaker. All these factors result in users developing a background level of stress, potentially explaining why most people feel exhausted after prolonged video conferencing [30] [27].

Fauville et al. [10, 11] theorized five nonverbal mechanisms that contribute to Zoom Fatigue. These include mirror anxiety, being physically trapped in the frustum of the camera, hyper-gaze, availability of nonverbal cues and the ability to interrupt discussion. They also argue how these factors can prove to be more stressful for women compared to men. The authors also developed a fatigue scale to quantitatively measure the negative effects of video conferencing. These negative effects can be significantly reduced by switching activities, forming small groups online and making lectures asynchronous [29].

Since the Covid-19 pandemic, and the wide scale adoption of video conferencing several studies and articles have attempted to model Zoom Fatigue [23, 24, 28] and employ techniques to mitigate this. Some simply present a number of steps that individuals can employ to combat tiredness during video conferencing, while others introduce several novel technological methodologies that organizations can incorporate in their systems to improve the Quality of Experience (QoE) during video conferencing [3, 14, 17]. QoE can be described as the degree of *annoyance* or *delight* of a consumer with a particular product or service. It is a holistic measurement of user experience and is a standard term used in the field of telecommunications [12].

3 USER STUDY

The aim of this user study is to understand the efficacy of visual communication through video conferencing in the presence of real world network latency and jitter. We use a game playing scenario in which participants play Charades with an actor using the Zoom video conferencing platform. The game of Charades is a guessing game where a player uses a set of actions to illustrate a phrase (for example, a movie title) and the other player(s) guess the enacted phrase. Charades is chosen because of its focus on visual communication necessary to study the effects of latency and jitter on visual communication in our study. We conducted a pilot study to enable us to formulate the design of the actual user study and evaluate empirical values for levels of injected latency and jitter, time to completion, number of game rounds, and movie title complexity.

3.1 Experimental Setup and Task

Our experiment was conducted during Covid lockdowns with 18 participants, aged between 18 and 22 years. 7 participants were female and 10 participants were male. All of the participants were familiar with the game of Charades. One of the authors *moderated* the sessions while another played the role of the *actor* for

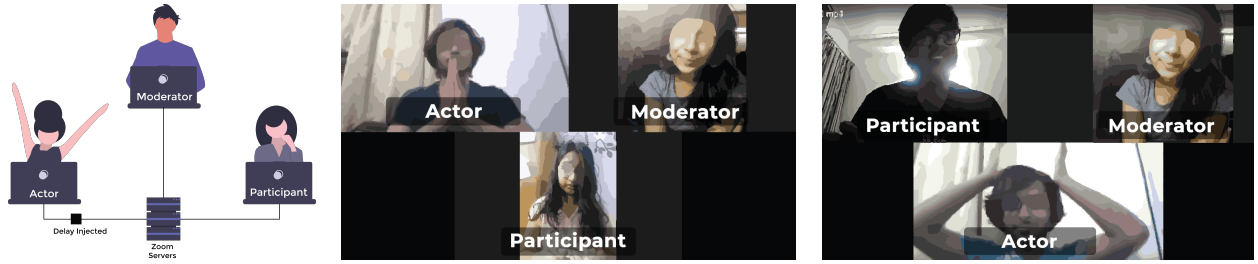


Figure 1: Experimental setup and snapshots from the user study.

the study (see Figure 1). We played four rounds of Charades with each participant and each round consisted of a single task (i.e., four tasks in total). Each task required participants to guess two movies, within a time limit of 6 minutes (maximum 3 minutes per movie). Each injected latency and jitter level was introduced in-between successive rounds. The first round was a calibration round with no injected latency and jitter. Between each round, a short survey was conducted to elucidate task difficulty and to pause before the next round. The actor used a predefined set of visual cues (actions) to enact the movies, to keep conditions consistent across all the participants and limit variability in visual cues used. The actor was not allowed to speak once the round commenced. Latency and jitter were injected at the actor's end, therefore a moderator also helped to ensure that the study proceeded smoothly.

3.1.1 Latency and Jitter Levels. To empirically determine levels of latency and jitter to use, we conducted a pilot study in which we increased levels of latency and jitter, based on a profiling of realistic network conditions. We selected levels where we were able to consistently see perceivable differences in performed actions from previous levels (these empirically derived levels are shown in Table 1). We found that Zoom calls faced frequent disconnections beyond 800ms of latency and 30% packet loss, hence, we used this as an upper threshold on the levels allowed in the study.

3.1.2 Movie Selection. We selected unrelated pairs of movies for each task balanced at equal levels of difficulty. It was important that the words in the movie title were actionable i.e. they could be enacted by the actor for participants to make a guess. To determine a viable set of movie titles, we first constructed a list of the top 25 movies watched worldwide. We first filtered these using a subjective evaluation of the words and if they were actionable or not. This allowed us to narrow down the list to eight candidate movies. We noted the list of words in each movie and ran the word analyzer algorithm [1] on each word to rank them objectively in terms of difficulty. The word analyzer algorithm gives a ranking metric as a function of word frequency. Frequent and familiar (less difficult) words are given a higher ranked value. We shortlisted a pair of movies for each round, balanced by their rank. There was no further contextual relationship between the movies chosen. We paired two movies with each other such that (see Table 2 for pairs):

- The number of effective words in the pair of movies through which an action can be demonstrated is 3 or 4.
- The average rank of the words within the actionable words set lie within the range of [7000-10000]. These are words that

participants will find moderately unfamiliar and therefore difficult to guess through the first action.

For each participant, we kept the order of injected latency and jitter levels the same (ascending) between rounds. However, we randomized the order of presentation of movie pairs for each participant (see Table 1 and Table 2). We took short breaks between rounds to conduct a brief survey and engaged participants in brief general conversation to minimise cumulative cognitive load over successive rounds.

Each participant performed 4 tasks of guessing 2 movies per task. The movies were same for all the participants but the order in which they were presented was randomised.

Table 1: Injected latency across tasks.

Task	Latency/Jitter Injected
1	No induced Latency and No Packet Loss
2	500 ms Latency and 15% Packet Loss [†]
3	650 ms Latency and 23% Packet Loss [†]
4	800 ms Latency and 30% Packet Loss [†]

[†]With 25% probability

3.2 Procedure

The authors secured ethics approval prior to undertaking this study from their Institutional Review Board (IRB). All participants were given a participant information sheet to inform them that they were taking part in a study on video conferencing. They were

Table 2: Pairs of movies with their average word analyzer ranks.

Movie Pair	Actionable Words	Rank
Chef; Fight Club	3 - (Chef, Fight, Club)	7580
The Godfather; Home Alone	4 - (God, father, home, Alone)	7074
Lord of the Rings; Sound of the Music	4 - (Lord, Ring, Sound, Music)	8953
Harry Potter and the Half-Blood Prince; Gravity	4 - (Harry, Half, Prince, Gravity)	9348

informed about the study by the authors and asked to fill-in a consent form confirming that they understood the study, the data collected, and their right to withdraw from the study at any point. Participants were then asked to join a video call over the Zoom video conferencing platform.

The actor first engaged with each participant in an informal conversation to check familiarity with the game. He confirmed with participants that no one else was consuming their available bandwidth. Following this, the actor explained the rules of the game for the purposes of the study. Participants were encouraged to take the lead and ask for visual cues to commence the round, then the actor began to act out the words of a movie at his own pace. We purposefully selected a set of visual actions that were not very easy to guess so as to engage the participant in the game for sufficiently long to observe any effects of injected network latency and jitter.

After each round participants were asked to fill-in a questionnaire based on the NASA-TLX [15] workload questionnaire. We also asked participants whether or not they were aware of the movie to help identify any biases. We took a short break after each round and used several conversational triggers (see Appendix A.2) to engage the participant in a brief unrelated conversation. This process was repeated four times. At the end of the experiment we conducted a semi-structured interview with each participant.

3.3 Hypotheses and Study Variables

Prior to conducting the study, we proposed the following hypotheses.

- **H1:** As network latency and jitter increases, the task becomes cognitively more demanding.
- **H2:** As network latency and jitter increases, the performance (guessing duration in < 180 sec) worsens in subsequent tasks.

Independent Variables. We identified the following independent variables in our user study

- *Network Latency/Jitter Injection.* The study comprised of 4 tasks with an increasing amount of injected latency and jitter in each task. We employed the NetEm [16] tool which builds upon the Linux traffic control facilities. The amount of network latency and packet loss injected for each successive task is shown in Table 1.
- *Actions for movies.* Each movie title was purposefully chosen to consist of actionable words such as Chef, Fight, Ring. The goal was to communicate the word to the participant through the actor's gestures alone and avoid any elicitation of the enacted word through memory.

Dependent Variables. The study design included the following quantitative and qualitative dependent variables

- *Task completion time (seconds).* We calculated the time each participant needed to guess the movies in each task. We placed a maximum allowed time of three minutes per movie for every task. Thus, the participant had a maximum total of 6 minutes guessing time per task round.
- *Correct movie guess.* We count the number of movies correctly guessed in each task by the participants.
- *NASA-TLX scores & User-Experience.* Participants were asked to note their experience along with their perception of the

task (How enjoyable was the task? 1-10). After completion of each task, we performed a short semi-structured interview asking the participants questions including "Why was this a fun activity for you?", "How was the overall experience?", "How did you perceive any glitches that you may have experienced during tasks?"

4 RESULTS

We analyze the quality of experience of the participants both objectively and subjectively. The number of movie guesses and the standard NASA-TLX questionnaire are used to obtain an objective score for the participant's experience. We capture the subjectivity of the experiment via semi-structured interviews. Our results are summarized below.

4.1 NASA-TLX Questionnaire

After each task, participants were asked to rate their perceived level of workload on the NASA-TLX standard scale [15]. This psychometric self-assessment scale consists of six dimensions: mental, physical, temporal workload and performance, effort, and frustration. The ratings for each part were collected on a 10-point Likert scale where 1 is very low and 10 is extremely high. Generally speaking participants experienced increasing levels of difficulty going from Task 1 to Task 4 with increased levels of network latency and jitter. We conducted Friedman ANOVA test on the NASA TLX results yielding p-values of $p_{mental} = 0.0828$, $p_{physical} = 0.0546$, $p_{temporal} = 0.2981$, $p_{performance} = 0.7128$, $p_{effort} = 0.7705$, and $p_{frustration} = 0.0789$. We observe that the test results have p-values greater than the α -value of 0.05. This suggests that the differences across various cognitive loads are not significant that can be partially attributed to our small sample size. We chose the Friedman test since it is a non-parametric test that makes no assumption on the normality of the distribution of the data. We discuss the individual aspects of the NASA-TLX ratings below. Figure 2 shows these statistics across the four tasks.

Physical workload. From our results, we can see that the overall physical load ratings have are reported to be lower than other factors. We speculate that this is due to the fact that participants had to only guess the word without putting much physical effort in performing the task and as such perceived it as a low physical workload task.

Performance workload. The trend for the performance load is the highest for task 1 (baseline latency and jitter) and remains around the same level. We attribute this to the reason that participants were very determined to correctly guess the movie within the stipulated time. P4 states "... I knew that action but time got over so quickly!" P10 states "... Why didn't you do this action for this word? I could have guessed it quickly." However, as the tasks progressed, the performance load seems to have remained fairly constant.

Temporal and Mental workload. The statistics for temporal workload and mental workload suggest a slight increase from task 1 to task 2 shown in Figure 2. Beyond task 2, there is a negligible change in subsequent tasks. This supports the study observations that participants adapted to latency and jitter adopting strategies to optimize their performance during the game and adapting to

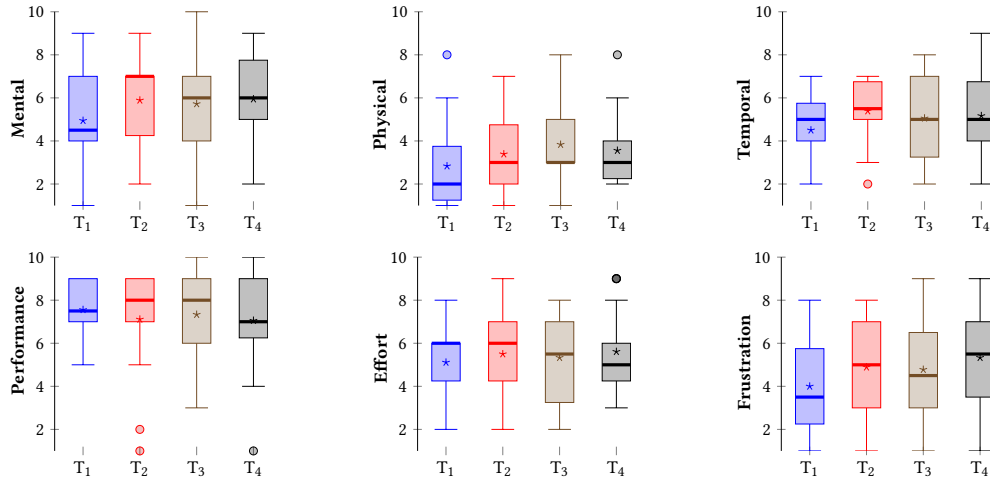


Figure 2: NASA Task Load Index ratings (self-assessment) across four tasks T₁, T₂, T₃, and T₄ (★ indicates mean). It can be observed that with increasing network latency and jitter, the mental workload and frustration of participants increased.

the pace of the game as the tasks progressed. We noticed that participants started moving their eye gaze away from the screen in later rounds with higher latency and jitter levels. After spending some time trying to understand the cues from the actor, participants stopped looking for further cues and chose to rely on the information they already have and started making guesses by recalling similar movies from their memory.

Frustration and Effort workload. The frustration and effort workload increased as participants progressed through tasks. 14 out of 18 participants expressed their concerns and their annoyance over latency at the end of task 4. The effort workload is self assessed by participants by asking them about their combined mental and physical effort.

4.2 User Performance

We show the results of performance of participants in guessing the movies in Figure 3. We kept a record of whether the user was able to guess the movie or not (or if partially guessed) for each movie enacted. The second movie guesses suggest a steady increase in failed cases across tasks, while no such pattern is observed in the first movie guesses. As previously mentioned, guessing the title of the movie brings into play some subjective considerations for example, the user may have watched the movie recently or may have guessed the whole movie by just guessing one word. We control for this subjectivity by probing any bias through a questionnaire.

5 DISCUSSION

Apart from the prominent factors of network latency and jitter, there were several other factors like enjoyment, and motivation of the participants that played a major role in our study. A straightforward expectation from such an experiment is that with increasing network latency and jitter, the performance of participants should deteriorate, however it is the interplay of other human factors such

as engagement and ability to adapt to network degradation that makes the outcome non-trivial. We summarize these factors below.

Success rate. The baseline task (Task 1) shows the maximum guesses of the movies without recall. However, as the tasks progressed, and consequently latency and jitter increased, participants required greater mental effort to guess the movie within the time limit of 3 minutes as indicated in section 4. In this case, they were more inclined to adopt a strategy of greater reliance on their own memory rather than the visual cues. This finding strongly suggests that if there is network latency and jitter in the system, participants will tend to use greater mental effort to interpret the other person’s visual communication and rely more on making predictions of what the other person is trying to communicate. This greater reliance on prediction based strategies could cause more errors in interpreting communication in complex tasks that rely on visual cues.

Fun factor. Overall, majority of participants indicated high levels of enjoyment. Maximum number of participants reported higher numbers - greater than 6 - on the Likert scale for each task. Even though some participants reported a lower score of 4 or 5 after task 2, there were more participants reporting a score of 10 beyond task 2 despite the injected network latency and jitter in the system. Participants who reported decreasing scores for enjoyment or low scores were either unable to guess the movie, or took longer than 1 minute to guess the movie. Despite guessing the movie, the latency in subsequent rounds troubled them more than participants who were able to guess in less than 1 minute. Such participants were also irritated at the end of the experiment and demonstrated increasing levels of stress as indicated by their NASA-TLX scores on mental workload and frustration levels.

Perceived network latency. At the end of every task, we asked the candidates if they perceived any network latency during the experiment. The participants reported the perceived latency on a Likert scale of 1-5, where 1 implies no network latency perceived and 5 implies network latency present for the entire task. Overall there is a clear shift in perception of network latency from task 1 to

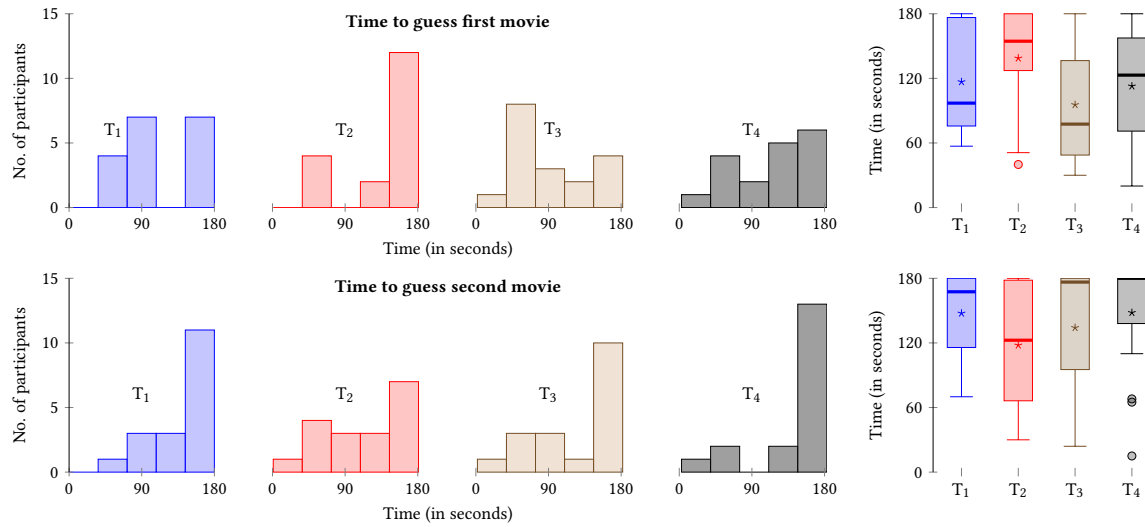


Figure 3: Histograms and statistics of movie guessing times (upper threshold - 180 sec.).

4. A noticeable change occurred from 14/18 participants reporting no latency in the system during task 1 to only 3/18 participants reporting the same in task 2. This number remains almost the same for rest of the tasks. Another observation is that after the no-latency round participants began to handle the additional injected latency and jitter and adapt to it in subsequent tasks.

6 CONCLUSION

In this study we presented the effect of network latency and jitter on visual task performance to assess the impact of network conditions on understanding non-verbal cues over video conferencing. These cues are important to collaboration in face-to-face communication and available in video but potentially disrupted by latency and jitter. Our observations suggest that prominent effects of network latency on participants arise from the mental workload and frustration during visual tasks as indicated by the NASA-TLX scores. An interesting effect is that majority of the participants in our study did adapt to the increasing levels of additional latency and jitter in the system throughout the whole experiment. Specifically, although the winning participants did experience an increased level of network latency and jitter in the later tasks, there was no significant performance loss due to degrading network quality. We attribute this to the high engagement level of the activity of playing the game and the ability of participants to adapt employing prediction while communicating. We also note that the frustration levels of all participants increased over time that indicates increased levels of fatigue owing to high mental workload. The visual nature of the task indicates a higher engagement level and consequently better performance in case of degraded network connectivity. In future work, we would like to investigate whether the same effects are present during other visual tasks involving video conferencing activities such as in formal meetings and technical discussions that may not be as engaging as playing a social game of Charades.

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A QUESTIONNAIRES

A.1 Post task interview (survey and NASA-TLX)

1. Round (1/2/3/4): _____
2. Were you able to guess the 1st movie?
 - ☐ Yes.
 - ☐ No.
3. Did you recollect the 1st movie from memory or you guessed it by the visual cues?
 - ☐ I guessed whatever I could, through the visual cues
 - ☐ I had heard/ watched / known about the movie previously, so could guess it easily
 - ☐ Both
 - ☐ Other: _____
4. Were you able to guess the 2nd movie?
 - ☐ Yes.
 - ☐ No.
5. Did you recollect the 2nd movie from memory or you guessed it by the visual cues?
 - ☐ I guessed whatever I could, through the visual cues
 - ☐ I had heard/ watched / known about the movie previously, so could guess it easily
 - ☐ Both

☐ Other: _____

6. How much mental and perceptual activity was required? Was the task easy (1) or demanding (10), simple (1) or complex (10)?
1 ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ 10
7. How much physical activity was required? Was the task easy (1) or demanding (10), slack (1) or strenuous (10)?
1 ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ 10
8. How much time pressure did you feel due to the pace at which the tasks or task elements occurred? Was the pace slow (1) or rapid (10)?
1 ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ 10
9. How successful were you in performing the task? How satisfied were you with your performance? Not successful at all (1), Successful (10)
1 ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ 10
10. How hard did you have to work (mentally and physically) to accomplish your level of performance? Easy (1), Complex (10)
1 ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ 10
11. How irritated, stressed, and annoyed (10) versus content, relaxed, and complacent (1) did you feel during the task?
1 ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ 10

A.2 Conversational triggers

1. What is your favourite movie?
2. Where would you like to go for a vacation?
3. How are you coping with the pandemic?
4. What is your favourite TV-show?
5. Which device do you prefer for video conferencing?
6. What genre of movies do you like?
7. Which video conferencing platform do you prefer?
8. Are you currently working from home?
9. What is the next item on your shopping wish-list?